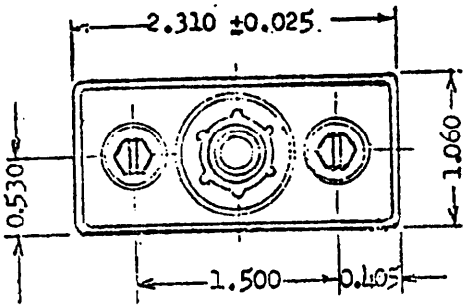
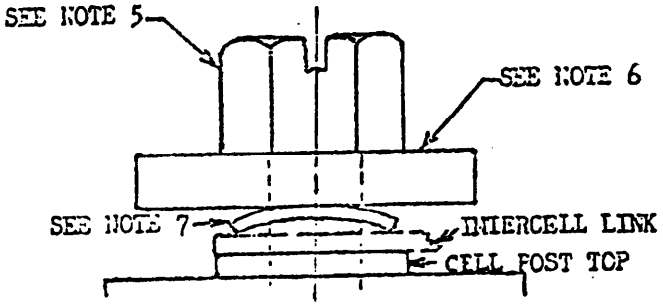
MIL-B-55118/5(EL)
29 November 1971

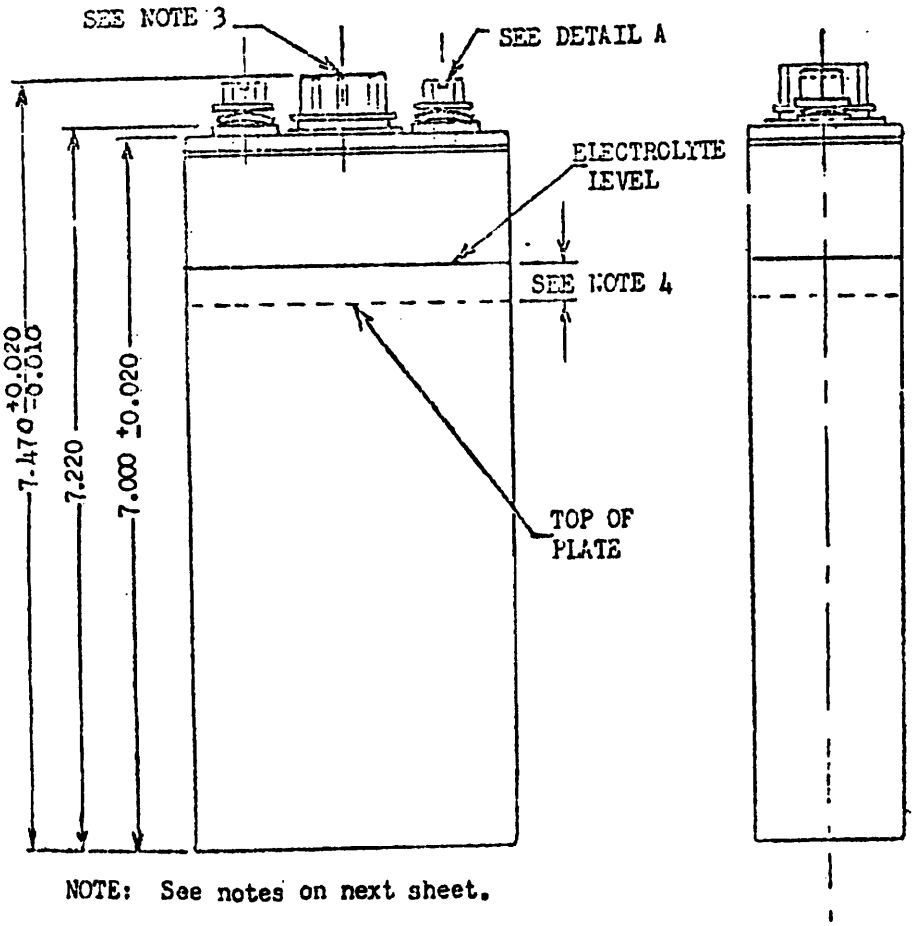
MILITARY SPECIFICATION SHEET
BATTERY, STORAGE BB-438/U

The complete requirements for procuring storage batteries described herein shall consist of this document and the issue in effect of Specification MIL-B-55118(EL).





DETAIL A



INCHES	MM
0.010	0.25
0.015	0.38
0.020	0.51
0.405	10.29
0.530	13.46
1.060	26.92
1.500	38.10
2.310	58.67
7.000	177.8
7.220	183.3
7.470	189.7

NOTE: See notes on next sheet.

FSC 6140

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NOTES (Applicable to foregoing drawing):

1. All dimensions shown on figure are in inches. Unless otherwise specified, tolerance is ± 0.015 in.

2. Metric equivalent shown on specification sheet are for general information only and are based upon 1 inch = 25.4 millimeters.

3. The standard bayonet type filler cap shall be Military Part No. MS75046-2.

4. An electrolyte level line should not be furnished on cell's container. The level of electrolyte shall be measurable by a glass tube. Level shall be adjusted with distilled water to 1/4 to 3/8 in. above the top of the plates aft full charge.

5. The slotted hex head screw shall be Military Part No. MS75043-2.

6. The flat washer on each terminal shall be Military Part No. MS75045-2.

7. The Belleville spring on each terminal shall be Military Part No. MS75044-2.

8. The 7.220 height dimension measures from cell bottom to the surface of the cell post top that contacts the bottom of the intercell link, as shown in Detail A.

9. The intercell link shown in Detail A is not furnished as part of the cell when cells are procured.

10. The head-space for the cell is defined as the space between plate tops and bottom of the cell's cover for electrolyte and gas. This space shall be at least twenty (20)% of the overall cell container volume.

11. For design feature purposes, this specification sheet takes precedence over procurement documents referenced herein.

12. Referenced documents shall be of the issue in effect on date of invitation for bid unless a specific exception is taken to same.

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BATTERY, STORAGE
BB-438/U
(CELL)

REQUIREMENTS:

Class: Vented nickel cadmium cell

Dimensions and configuration: See figure and notes.

Weight (max.): 1.75 pounds.

Terminals: Two slotted hex hd screws, each complete with Belleville spring and flat washer per figure and notes.

Container: Polyamide, ABS or equal.

Head space: 20% minimum per Note 10.

Filler cap: Std. per Note 3.

Vent release pressure: 6.0 ± 4.0 psig.

Nominal voltage: 1.2 volts.

Final voltage: 1.0 volt.

Rated capacity, (5 hr. rate): 20.0 amp. hrs.

Normal charge: 0.33 C_5 to 1.50 volts, then 0.17 C_5 for 2.0 hrs.

Normal discharge rate: 0.20 C_5.

0.17 C_5 rate: 3.4 amps.

0.20 C_5 rate: 4.0 amps.

0.23 C_5 rate: 4.6 amps.

0.33 C_5 rate: 6.6 amps.

3.0 C_5 rate: 60.0 amps.

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BATTERY, STOPAGE
(CELL)
BB-438/U

First Article Inspections	MIL-B-55118 Inspection Paragraph	Requirement
Visual and mechanical examination	4.7.1	Table IX
Initial full capacity. Discharged (upright)	4.7.14.1	20.0 AH @ 0.20 C_5
Discharged (on side)	4.7.3.1	20.0 AH @ 0.20 C_5
Cycle life	4.7.15	1008 cycles to 14.0 AH
Potassium carbonate	4.7.7	Less than 60 grams per liter
Retention of charge I (14 days)	4.7.13.1	18.0 AH @ 0.20 C_5
Retention of charge II (4 days)	4.7.13.2	8.0 AH @ 0.20 C_5
High Rate Discharge	4.7.14.4	15.0 AH @ 3.0 C_5
Low temperature	4.7.14.5	12.0 AH @ 0.20 C_5
Temperature cycling	4.7.10	No physical defects per Table IX
Full capacity (upright)	4.7.14.3	20.0 AH @ 0.20 C_5
Mechanical shock	4.7.11	No physical defects per Table IX
Full capacity (upright)	4.7.14.3	20.0 AH @ 0.20 C_5
Vibration	4.7.12	No physical defects per Table IX
Full capacity (upright)	4.7.14.3	20.0 AH @ 0.20 C_5
Internal pressure	4.7.2	No leakage
Electrical leakage	4.7.9	More than 50 megohms

Quality conformance inspection:

It encompasses same inspections as first article except cycle life shall be 504 cycles to 17.0 AH.

Sampling: See latest issue of MIL-B-55118(EL).

Custodian:
Army-EL

Preparing Activity
Army-EL

Project No. 6140-A303

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-B-55118/5(EL) MILITARY SPECIFICATION SHEET BATTERY, STORAGE BB-438/U

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

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